

Project Executable Files

Date	2 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	No

6	Deployed application URL (if hosted)	No
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```
▼ OptiCrop
  ▼ models
    ≡ crop_recommendation_model.pkl
    ≡ scaler.pkl
  ▼ static
    > css
    > img
    > js
  ▼ templates
    <> index.html
  ▼ venv
    > etc
    > Include
    > Lib
    > Scripts
    > share
    ◆ .gitignore
    ⚙ pyvenv.cfg
  ⚙ .env
  ◆ .gitignore
  + app.py
  + data_pipeline.py
  📄 dataset.csv
  + download_images.py
  + generate_dataset.py
  + generate_images.py
  ⓘ README.md
  ≡ requirements.txt
  + train_model.py
  > OptiCrop_New
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://smart-agriculture-6ois.onrender.com

Login Credentials (Demo)	Email- mannjain4885@gmail.com Logged in with mail
Platform / Hosting Provider	Render
Repository Link	https://github.com/mann-jain143/Smart--Agriculture
Demo Video Link	https://drive.google.com/file/d/1BXr_qBVO3tTq-BlytZNS7w7f5RMVla6-/view?usp=sharing

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

1. Download or clone the project repository.
2. Open the project folder in Visual Studio Code.
3. Create a virtual environment (optional).
4. Install the required dependencies using:

```
</> Bash
pip install -r requirements.txt
```

5. Run the Flask application:

```
python app.py
```

6. Open a web browser and visit:

```
http://127.0.0.1:5000/
```

7. Enter the required soil and weather parameters.
8. Click **Predict Crop** to view the recommended crop and confidence analysis.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Application currently runs on a local Flask server	Can be deployed on Render for public access
2	Predictions depend on user-provided input accuracy	Use recent soil testing and weather data
3	Limited to crops available in the training dataset	Future versions will include additional crop varieties